Step 2: Two-Sample Comparison (2023 vs 2024)

Danish Housing Prices · Mathematical Statistics

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```
library(tidyverse)
dk_villa <- readRDS("../../data/dk_villa.rds") %>%
  mutate(year = as.integer(substr(date, 1, 4)))

y2023 <- dk_villa$log_price[dk_villa$year == 2023]
y2024 <- dk_villa$log_price[dk_villa$year == 2024]

cat("n_2023 =", length(y2023), "| n_2024 =", length(y2024), "\n")
```

```
n_2023 = 225 | n_2024 = 177
```

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1 Motivation

Danish nominal interest rates rose sharply from 2022 onward. We ask: does this translate into a statistically detectable difference in mean log-price between 2023 and 2024 Copenhagen Villas?

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2 Statistical Model

$$Y_{ij} \sim \mathcal{N}(\mu_j, \sigma^2), \quad j \in \{2023, 2024\}, \quad i = 1, \dots, n_j \quad \text{(independent)}$$

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3 Variance Homogeneity — F-test

The course (§2, slides 4) prescribes the **F-test** for exactly $m = 2$ variance estimates:

$$H_{0\sigma^2} : \sigma^2_{2023} = \sigma^2_{2024}$$

$$F = \frac{S^2_{2023}}{S^2_{2024}} \sim F(n_1 - 1,\ n_2 - 1) \quad \text{under } H_{0\sigma^2}$$

```r
cat("Per-group SDs:\n")
cat("  2023:", round(sd(y2023), 4), "\n")
cat("  2024:", round(sd(y2024), 4), "\n\n")

vt <- var.test(y2023, y2024)
print(vt)

use_equal_var <- vt$p.value > 0.05
cat("\nConclusion: using equal variances =", use_equal_var, "\n")
```

```
Per-group SDs:
  2023: 0.2983
  2024: 0.2869


	F test to compare two variances

data:  y2023 and y2024
F = 1.0809, num df = 224, denom df = 176, p-value = 0.5898
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.8149994 1.4268422
sample estimates:
ratio of variances 
          1.080939 


Conclusion: using equal variances = TRUE
```

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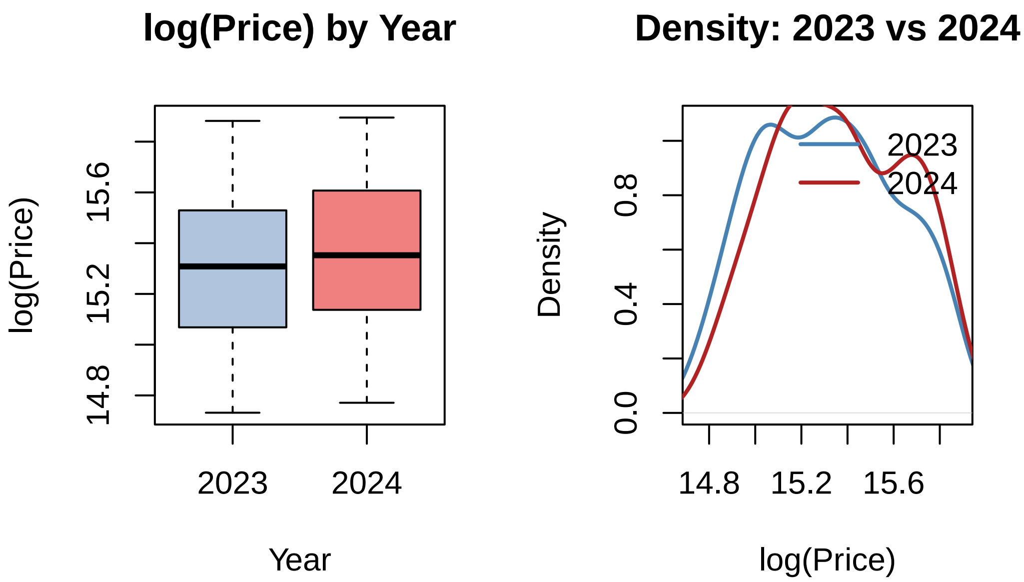
4 Exploratory Plots

```
par(mfrow = c(1, 2))

boxplot(log_price ~ year, data = dk_villa,
        col  = c("lightsteelblue", "lightcoral"),
        main = "log(Price) by Year",
        xlab = "Year", ylab = "log(Price)")

plot(density(y2023), col = "steelblue", lwd = 2,
     xlim = range(dk_villa$log_price),
     main = "Density: 2023 vs 2024", xlab = "log(Price)")
lines(density(y2024), col = "firebrick", lwd = 2)
legend("topright", legend = c("2023","2024"),
       col = c("steelblue","firebrick"), lwd = 2, bty = "n")
```




```
par(mfrow = c(1, 1))
```

5 Two-Sample t-Test

5.1 Hypothesis

$$H_0 : \mu_{2023} = \mu_{2024} \quad \text{vs} \quad H_1 : \mu_{2023} \neq \mu_{2024}$$

5.2 Test statistic (pooled variance)

$$T = \frac{\bar{Y}_1 - \bar{Y}_2}{S_p\sqrt{1/n_1 + 1/n_2}} \sim t(n_1 + n_2 - 2) \quad \text{where } S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

```r
n1  <- length(y2023);  n2  <- length(y2024)
sp2 <- ((n1-1)*var(y2023) + (n2-1)*var(y2024)) / (n1 + n2 - 2)
df  <- n1 + n2 - 2

t_stat <- (mean(y2023) - mean(y2024)) / sqrt(sp2 * (1/n1 + 1/n2))
p_val  <- 2 * pt(-abs(t_stat), df = df)
ci     <- (mean(y2023) - mean(y2024)) +
           c(-1, 1) * qt(0.975, df) * sqrt(sp2 * (1/n1 + 1/n2))

cat("mean_2023   =", round(mean(y2023), 4), "\n")
cat("mean_2024   =", round(mean(y2024), 4), "\n")
cat("sp          =", round(sqrt(sp2),   4), "\n")
cat("t-statistic =", round(t_stat, 4), "on", df, "df\n")
cat("p-value     =", round(p_val,  6), "\n")
cat("95% CI for (mu_2023 - mu_2024): [", round(ci, 4), "]\n")
cat("Back-transformed ratio CI: [",
    round(exp(ci), 4), "]\n")
```

```
mean_2023   = 15.3119
mean_2024   = 15.3637
sp          = 0.2934
t-statistic = -1.7576 on 400 df
p-value     = 0.079587
95% CI for (mu_2023 - mu_2024): [ -0.1098 0.0061 ]
Back-transformed ratio CI: [ 0.8961 1.0062 ]
```

```r
t.test(y2023, y2024, var.equal = use_equal_var)
```

```

	Two Sample t-test

data:  y2023 and y2024
```

```
t = -1.7576, df = 400, p-value = 0.07959
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.109752113  0.006141342
sample estimates:
mean of x mean of y
 15.31186  15.36366
```

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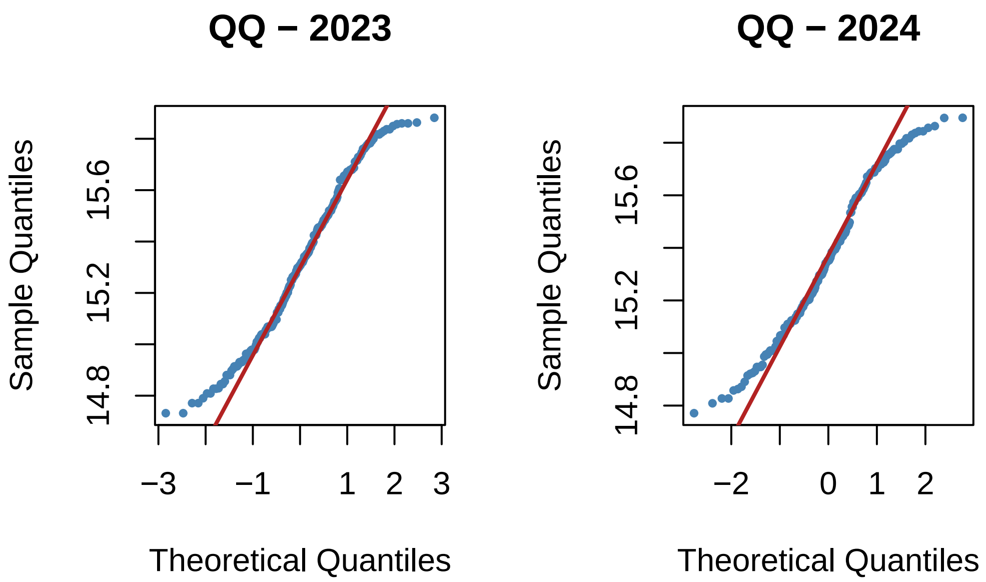
6 Normality Check per Group

```r
par(mfrow = c(1, 2))

qqnorm(y2023, main = "QQ - 2023",
       pch = 16, cex = 0.6, col = "steelblue")
qqline(y2023, col = "firebrick", lwd = 2)

qqnorm(y2024, main = "QQ - 2024",
       pch = 16, cex = 0.6, col = "steelblue")
qqline(y2024, col = "firebrick", lwd = 2)
```



```r
par(mfrow = c(1, 1))
```

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7 Interpretation

The back-transformed CI for $(\mu_{2023} - \mu_{2024})$ is on the log scale. Exponentiating gives a ratio: $\exp(\bar{Y}_{2023} - \bar{Y}_{2024})$ is the multiplicative factor by which the 2023 median price exceeds the 2024 median. A value above 1 means prices were higher in 2023; below 1 means lower.